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SYSTEMS AND METHODS FOR POWER TOPOLOGY MAPPING

Abstract

An information handling system may include a processor and a management agent configured to collect input power information and input voltage information associated with a plurality of power supply units, collect output power information and output voltage information associated with a plurality of outlets, and, for each outlet of the plurality of outlets, determine a power difference between such outlet and each of the plurality of power supply units and map such outlet to a power supply unit of the plurality of power supply units for which the power difference between such outlet and such power supply unit is smallest among all of the power differences between such outlet and each of the plurality of power supply units.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates in general to information handling systems, and more particularly to systems and methods for mapping power topology in a data center environment.

BACKGROUND

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

[0003] In a data center comprising multiple information handling systems, there is often a desire to gather configuration and status information, and create maps from this information to define a complete power and information technology equipment infrastructure or topology of the data center. Having a map of such topology may assist in data center management, allowing removal of guess work in determining a data center's power needs, reclamation of trapped power, and avoiding unscheduled downtime.

[0004] Such configuration and status information may include at least two types of data. One type of data is data regarding individual information handling system hardware configuration, such as computing load and power consumption, etc., which may be used to optimize system resource mapping. Another type of data is a physical power mapping between one or more power distribution units and individual power supply units, which may be used for balancing loads on each alternating current phase in order to optimize power sourcing. Existing approaches to such mapping typically involve manual entry of mapping the server rack location and its corresponding power distribution unit and power distribution outlet in a table or other data structure. Such manual data entry may be costly, time consuming, and prone to error.

SUMMARY

[0005] In accordance with the teachings of the present disclosure, the disadvantages and problems associated with mapping of a power topology may be reduced or eliminated.

[0006] In accordance with embodiments of the present disclosure, an information handling system may include a processor and a management agent configured to collect input power information and input voltage information associated with a plurality of power supply units, collect output power information and output voltage information associated with a plurality of outlets, and, for each outlet of the plurality of outlets, determine a power difference between such outlet and each of the plurality of power supply units and map such outlet to a power supply unit of the plurality of power supply units for which the power difference between such outlet and such power supply unit is smallest among all of the power differences between such outlet and each of the plurality of power supply units.

[0007] In accordance with these and other embodiments of the present disclosure, a method may include collecting input power information and input voltage information associated with a plurality of power supply units, collecting output power information and output voltage information associated with a plurality of outlets, and for each outlet of the plurality of outlets,

determining a power difference between such outlet and each of the plurality of power supply units, and mapping such outlet to a power supply unit of the plurality of power supply units for which the power difference between such outlet and such power supply unit is smallest among all of the power differences between such outlet and each of the plurality of power supply units.
[0008] In accordance with these and other embodiments of the present disclosure, an article of manufacture may include a non-transitory computer-readable medium and computer-executable instructions carried on the computer-readable medium, the instructions readable by a processor, the instructions, when read and executed, for causing the processor to: collect input power information and input voltage information associated with a plurality of power supply units; collect output power information and output voltage information associated with a plurality of outlets; and for each outlet of the plurality of outlets, determine a power difference between such outlet and each of the plurality of power supply units and map such outlet to a power supply unit of the plurality of power supply units for which the power difference between such outlet and such power supply unit is smallest among all of the power differences between such outlet and each of the plurality of power supply units.
[0009] Technical advantages of the present disclosure may be readily apparent to one skilled in the art from the figures, description and claims included herein. The objects and advantages of the embodiments will be realized and achieved at least by the elements, features, and combinations particularly pointed out in the claims.
[0010] It is to be understood that both the foregoing general description and the following detailed description are examples and explanatory and are not restrictive of the claims set forth in this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0011] A more complete understanding of the present embodiments and advantages thereof may be acquired by referring to the following description taken in conjunction with the accompanying drawings, in which like reference numbers indicate like features, and wherein:
[0012] FIG. **1** illustrates a block diagram of an example system, in accordance with embodiments of the present disclosure;
[0013] FIG. **2** illustrates a block diagram of an example information handling system, in accordance with embodiments of the present disclosure;
[0014] FIG. **3** illustrates a block diagram of an example power train, in accordance with embodiments of the present disclosure;
[0015] FIG. **4** illustrates a block diagram of an example power distribution unit, in accordance with embodiments of the present disclosure;
[0016] FIG. **5** illustrates a block diagram of an example system manager, in accordance with embodiments of the present disclosure;
[0017] FIG. **6** illustrates a flow chart of an example method for mapping a power supply unit to a power distribution unit and outlet, in accordance with embodiments of the present disclosure; and
[0018] FIG. **7** illustrates a flow chart of an example method for verifying a mapping of a power supply unit to a power distribution unit and outlet, in accordance with embodiments of the present disclosure.
DETAILED DESCRIPTION
[0019] Preferred embodiments and their advantages are best understood by reference to FIGS. **1** through **7**, wherein like numbers are used to indicate like and corresponding parts.
[0020] For the purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit,

receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, an information handling system may be a personal computer, a personal data assistant (PDA), a consumer electronic device, a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include memory, one or more processing resources such as a central processing unit (CPU) or hardware or software control logic. Additional components of the information handling system may include one or more storage devices, one or more communications ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. The information handling system may also include one or more buses operable to transmit communication between the various hardware components.

[0021] For the purposes of this disclosure, computer-readable media may include any instrumentality or aggregation of instrumentalities that may retain data and/or instructions for a period of time. Computer-readable media may include, without limitation, storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk), a sequential access storage device (e.g., a tape disk drive), compact disk, CD-ROM, DVD, random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), and/or flash memory; as well as communications media such as wires, optical fibers, microwaves, radio waves, and other electromagnetic and/or optical carriers; and/or any combination of the foregoing.

[0022] For the purposes of this disclosure, information handling resources may broadly refer to any component system, device or apparatus of an information handling system, including without limitation processors, service processors, basic input/output systems (BIOSs), buses, memories, I/O devices and/or interfaces, storage resources, network interfaces, motherboards, power supplies, air movers (e.g., fans and blowers) and/or any other components and/or elements of an information handling system.

[0023] FIG. **1** illustrates a block diagram of an example system **100** which may represent at least a portion of components present in a data center environment. As shown in FIG. **1**, system **100** may comprise a plurality of information handling systems **102**, one or more power distribution units **104**, a system manager **106**, and a network **108** communicatively coupled to the information handling systems **102**, power distribution units **104**, and system manager **106**.

[0024] In some embodiments, each information handling system **102** may comprise a server. As shown in FIG. **1**, each information handling system **102** may include at least one power supply unit (PSU) **110** configured to receive electrical energy from a corresponding power outlet of power distribution unit **104** in order to provide power to components of information handling system **102**. Although FIG. **1** depicts each information handling system **102** having two PSUs **110**, an information handling system **102** may include any suitable number of PSUs **110** each of which may be supplied electrical energy from a corresponding outlet of a power distribution unit **104**.

[0025] Although FIG. **1** depicts system **100** having two power distribution units **104**, system **100** may include any suitable number of power distribution units **104**. In embodiments including a plurality of power distribution units **104**, some power distribution units **104** may receive a different alternating current source as shown by “AC FEED 1” and “AC FEED 2” in FIG. **1**.

[0026] Also as shown in FIG. **1**, a system manager **106** for monitoring, control, and management of power distribution units **104** and information handling systems **102** may be coupled via network **108** to power distribution units **104** and information handling systems **102** (e.g., via Ethernet).

[0027] Network **108** may be a network and/or fabric configured to couple system manager **106** to information handling systems **102**, power distribution units **104**, and/or one or more other information handling systems. In these and other embodiments, network **108** may include a communication infrastructure, which provides physical connections, and a management layer, which organizes the physical connections and information handling systems communicatively coupled to network **108**. Network **108** may be implemented as, or may be a part of, a storage area network (SAN), personal area network (PAN), local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a wireless local area network (WLAN), a virtual private network (VPN), an intranet, the Internet or any other appropriate architecture or system that facilitates the communication of signals, data and/or messages (generally referred to as data). Network **108** may transmit data via wireless transmissions and/or wire-line transmissions using any storage and/or communication protocol, including without limitation, Fibre Channel, Frame Relay, Asynchronous Transfer Mode (ATM), Internet protocol (IP), other packet-based protocol, small computer system interface (SCSI), Internet SCSI (iSCSI), Serial Attached SCSI (SAS) or any other transport that operates with the SCSI protocol, advanced technology attachment (ATA), serial ATA (SATA), advanced technology attachment packet interface (ATAPI), serial storage architecture (SSA), integrated drive electronics (IDE), and/or any combination thereof. Network **108** and its various components may be implemented using hardware, software, or any combination thereof.

[0028] FIG. **2** illustrates a block diagram of an example of an information handling system **102**. As depicted, information handling system **102** may include PSU **110**, a motherboard **201**, and one or more other information handling resources.

[0029] Motherboard **201** may include a circuit board configured to provide structural support for one or more information handling resources of information handling system **102** and/or electrically couple one or more of such information handling resources to each other and/or to other electric or electronic components external to information handling system **102**. As shown in FIG. **2**, motherboard **201** may include a processor **203**, memory **204**, a management controller **206**, and one or more other information handling resources.

[0030] Processor **203** may comprise any system, device, or apparatus operable to interpret and/or execute program instructions and/or process data, and may include, without limitation a microprocessor, microcontroller, digital signal processor (DSP), application specific integrated circuit (ASIC), or any other digital or analog circuitry configured to interpret and/or execute program instructions and/or process data. In some embodiments, processor **203** may interpret and/or execute program instructions and/or process data stored in memory **204** and/or another component of information handling system **102**.

[0031] Memory **204** may be communicatively coupled to processor **203** and may comprise any system, device, or apparatus operable to retain program instructions or data for a period of time. Memory **204** may comprise random access memory (RAM), electrically erasable programmable read-only memory (EEPROM), a PCMCIA card, flash memory, magnetic storage, opto-magnetic storage, or any suitable selection and/or array of volatile or non-volatile memory that retains data after power to information handling system **102** is turned off.

[0032] Management controller **206** may be configured to provide out-of-band management facilities for management of information handling system **102**. Such management may be made by management controller **206** even if information handling system **102** is powered off or powered to a standby state. Management controller **206** may include a processor, memory, out-of-band network interface **208** separate from and physically isolated from an in-band network interface of information handling system **102**, and/or other embedded information handling resources. In certain embodiments, management controller **206** may include or may be an integral part of a baseboard management controller (BMC) or a remote access controller (e.g., a Dell Remote Access Controller or Integrated Dell Remote Access Controller). In other embodiments, management controller **206** may include or may be an integral part of a chassis management controller (CMC). In some embodiments, management controller **206** may be configured to communicate with PSU **110** to communicate control and/or telemetry data between the two.

[0033] Network interface **208** may comprise any suitable system, apparatus, or device operable to serve as an interface between management controller **206** and another information handling system (e.g., system manager **106**) and/or a network (e.g., network **108**). Network interface **208** may enable management controller **206** to communicate using any suitable transmission protocol and/or standard. In some embodiments, network interface **206** may be configured to communicate with other information handling systems via one or more protocols or standards discussed above with respect to network **108**. In these and other embodiments, network interface **208** may comprise a network interface card, or "NIC."

[0034] Generally speaking, PSU **110** may include any system, device, or apparatus configured to supply electrical current to one or more information handling resources of information handling system **102**. As shown in FIG. **2**, PSU **110** may include one or more microcontroller units (MCUs) **212** and a power train **214**.

[0035] An MCU **212** may comprise a microprocessor, DSP, ASIC, FPGA, EEPROM, or any combination thereof, or any other device, system, or apparatus for controlling operation of its associated PSU **110**. As such, an MCU **212** may comprise firmware, logic, and/or data for controlling functionality of such PSU **110**. In some embodiments, MCU **212** may also be configured to meter one or more parameters (e.g., voltage, current, input power, etc.) indicative of electrical energy received by power train **214** from a power distribution unit **104**. In some embodiments, MCU **212** may be configured to communicate such metered information to system manager **106** via management controller **206** and network interface **208**.

[0036] Power train **214** may include any suitable system, device, or apparatus for converting electrical energy received from power distribution unit **104** (e.g., a 120-volt alternating current voltage waveform) into electrical energy usable to information handling resources of information handling system **102** (e.g., 12-volt direct current voltage source). Selected components of an example power train **214** are shown in FIG. **3** below.

[0037] In addition to motherboard **201**, processor **203**, memory **204**, management controller **206**, network interface **208**, and PSU **110**, information handling system **102** may include one or more other information handling resources.

[0038] FIG. **3** illustrates a block diagram of an example power train **214**, in accordance with embodiments of the present disclosure. As shown in FIG. **3**, power train **214** may include two converter stages: a rectifier/power factor correction (PFC) stage **302**, a DC/DC converter stage **304**, a bulk capacitor **306** coupled between an output of rectifier/PFC stage **302** and an input of DC/DC converter stage **304** and an output capacitor **308** coupled to an output of DC/DC converter stage **304**.

[0039] Rectifier/PFC stage **302** may be configured to, based on an input current i.sub.IN, a sinusoidal voltage source v.sub.IN, and a bulk capacitor voltage V.sub.BULK, shape the input current i.sub.IN to have a sinusoidal waveform in-phase with the source voltage v.sub.IN and to generate regulated DC bus voltage V.sub.BULK on bulk capacitor **306**.

[0040] DC/DC converter stage **304** may convert bulk capacitor voltage V.sub.BULK to a DC output voltage V.sub.OUT on output capacitor **308** which may be provided to a load (e.g., to information handling resources of information handling system **102** in order to power such information handling resources). In some embodiments, DC/DC converter stage **304** may be implemented as a converter which converts a higher DC voltage (e.g., 400 V) into a lower DC voltage (e.g., 12 V).

[0041] FIG. **4** illustrates a block diagram of an example power distribution unit **104**, in accordance with embodiments of the present disclosure. As shown in FIG. **4**, power distribution unit **104** may be an "intelligent" power distribution unit **104** comprising a controller **402**, a plurality of outlets **404**, a plurality of meters **406**, and a network interface **408**.

[0042] Controller **402** may comprise any system, device, or apparatus operable to monitor and/or control operation of power distribution unit **104**, including control of operation of outlets **404** and the distribution of power thereto, and the reading and/or processing of information from meters **406**. Controller **402** may include, without limitation a microprocessor, microcontroller, digital signal processor (DSP), application specific integrated circuit (ASIC), or any other digital or analog circuitry configured to interpret and/or execute program instructions and/or process data. In some embodiments, controller **402** may interpret and/or execute program instructions (e.g., firmware) and/or process data stored in computer-readable media accessible to controller **402**.

[0043] In operation, power distribution unit **104** may distribute electrical energy received from a power source (e.g., a nominally 60 Hz/110 V line voltage in the United States of America or a nominally 50 Hz/220 V line voltage in Europe) to one or more outlets **404**. Each outlet **404** may comprise a suitable electrical connector (e.g., a female electrical connector) for receiving a plug or other male connector of a device (e.g., a PSU **110** of an information handling system **102**) in order to deliver electrical energy to such device.

[0044] Each meter **406** may be associated with a corresponding outlet **404** (e.g., in a one-to-one correspondence) and may include any system, device, or apparatus configured to meter one or more parameters (e.g., voltage, current, output power, etc.) indicative of electrical energy delivered from the corresponding outlet **404**. In some embodiments, a meter **406** may be configured to communicate such metering information to system manager **106** via controller **402** and network interface **408**.

[0045] Network interface **408** may comprise any suitable system, apparatus, or device operable to serve as an interface between power distribution unit **104** and an information handling system (e.g., system manager **106**) and/or a network (e.g., network **108**). Network interface **408** may enable power distribution unit **104** to communicate using any suitable transmission protocol and/or standard. In some embodiments, power distribution unit **104** may be configured to communicate with other information handling systems (including, without limitation, system manager **106**) via one or more protocols or standards discussed above with respect to network **108**. In these and other embodiments, network interface **408** may comprise a NIC.

[0046] FIG. **5** illustrates a block diagram of an example system manager **106**, in accordance with embodiments of the present disclosure. As described above, system manager **106** may be configured to monitor, control, and manage power distribution unit **104** and information handling systems **102**. In some embodiments, system manager **106** may comprise a "top or rack" switch for a server rack comprising information handling systems **102**. In other embodiments, system manager **106** may comprise a personal computer, such as a laptop, notebook, or desktop computer. In yet other embodiments, system manager **106** may be a mobile device sized and shaped to be readily transported and carried on a person of a user of information handling system **102** (e.g., a smart phone, a tablet computing device, a handheld computing device, a personal digital assistant, etc.). As shown in FIG. **5**, system manager **106** may comprise a processor **503**, a memory **504**, and a network interface **508**.

[0047] Processor **503** may comprise any system, device, or apparatus operable to interpret and/or execute program instructions and/or process data, and may include, without limitation a microprocessor, microcontroller, digital signal processor (DSP), application specific integrated circuit (ASIC), or any other digital or analog circuitry configured to interpret and/or execute program instructions and/or process data. In some embodiments, processor **503** may interpret and/or execute program instructions and/or process data stored in memory **504** and/or another component of system manager **106**.

[0048] Memory **504** may be communicatively coupled to processor **503** and may comprise any system, device, or apparatus operable to retain program instructions or data for a period of time. Memory **504** may comprise random access memory (RAM), electrically erasable programmable read-only memory (EEPROM), a PCMCIA card, flash memory, magnetic storage, opto-magnetic storage, or any suitable selection and/or array of volatile or non-volatile memory that retains data after power to system manager **106** is turned off. As shown in FIG. **5**, memory **504** may have stored thereon a management agent **506**. Management agent **506** may include a program of instructions that may be read and executed by processor **503** in order to carry out the management functionality

of system manager **106**, as such functionality is described elsewhere in this disclosure.
[0049] Network interface **508** may comprise any suitable system, apparatus, or device operable to serve as an interface between system manager **106** and one or more information handling systems (e.g., information handling systems **102**), other networked devices (e.g., power distribution unit **104**) and/or a network (e.g., network **108**). Network interface **508** may enable system manager **106** to communicate using any suitable transmission protocol and/or standard. In some embodiments, system manager **106** may be configured to communicate with other information handling systems via one or more protocols or standards discussed above with respect to network **108**. In these and other embodiments, network interface **508** may comprise a NIC.
[0050] In operation, management agent **506** may group or “zone” information handling systems **102** together as a set. For example, management agent **506** may group information handling systems **102** together which are coupled to the same top of rack switch, for example by using an Internet Protocol address table (e.g., stored in memory **504** of system manager **106** or a top of rack switch) which may indicate those information handling systems located in the same server rack. Information handling systems **102** coupled to the same top of rack switch are likely to be powered from the same set of power distribution units **104**.
[0051] After zoning information handling systems **102** likely to be powered from the same set of power distribution units **104**, management agent **506** may automatically map individual outlets **404** of power distribution units **104** to the respective individual PSUs **110** coupled with (i.e., plugged into) such outlets **404**. To perform such automatic mapping, management agent **506** may sample instantaneous input power and voltage metrics from all PSUs **110** and sample instantaneous output power and voltage metrics from all outlets **404** for a defined period of time (e.g., every 10 seconds for two minutes). In some embodiments, management agent **506** may discard from the mapping process outlets **404** and PSUs **110** that report zero readings for voltage and/or power. Management agent **506** may further group PSUs **110** and outlets **404** having similar voltage values together. In addition, management agent **506** may for each outlet **404**, determine a power difference between the output power of such outlet **404** and the input power of every PSU **110**. Further, management agent **506** may map each outlet **404** to the PSU **110** having, out of all of the PSUs **110**, the smallest power difference between the output power of such outlet **404** and the input power of such PSU **110**.
[0052] After mapping each outlet **404** to a respective PSU **110**, management agent **506** may verify the mappings. To perform such verification for a mapping of a particular PSU **110** of one information handling system **102** to a particular outlet **404**, management agent **506** may cause all other information handling systems **102** except for the particular information handling system **102** to be power capped. Management agent **506** may then cause power consumption on the particular information handling system **102** to momentarily change and determine if a meter **406** associated with the outlet **404** mapped to the information handling system **102** reports a similar change in power delivery. If the change in power delivery at the outlet **404** is similar to the change in power consumption at the information handling system **102**, then the mapping is verified. Such verification process may be repeated for each mapping of outlets **404** to PSUs **110**.
[0053] FIG. **6** illustrates a flow chart of an example method **600** for mapping PSUs **110** to power distribution units **104** and outlets **404** thereof, in accordance with embodiments of the present disclosure. According to certain embodiments, method **600** may begin at step **602**. As noted above, teachings of the present disclosure may be implemented in a variety of configurations of system **100**. As such, the preferred initialization point for method **600** and the order of the steps comprising method **600** may depend on the implementation chosen.
[0054] At step **602**, management agent **506** may collect instantaneous input power and voltage reported by MCUs **212** of PSUs **110** a number of t times at regular intervals. A variable IP.sub.t.sup.n may define an input power of a PSU PSU.sub.N at a time instant t, and a variable IV.sub.t.sup.n may define an input voltage of such PSU PSU.sub.N at time instant t.

[0055] At step **604**, management agent **506** may collect instantaneous output power and voltage reported by meters **406** of outlets **404** at the same number of t times. A variable OP.sub.t.sup.n may define an output power of an outlet PDUoutlet.sub.N at time instant t, and a variable OV.sub.t.sup.n may define an output voltage of such outlet PDUoutlet.sub.N at time instant t.
[0056] At step **606**, management agent **506** may discard from consideration any PSUs **110** and outlets **404** reporting zero values for power and/or voltage.
[0057] At step **608**, management agent **506** may create voltage groups of PSUs **110** and outlets **404** having similar average voltage values. For example, in some embodiments, management agent **506** may use K-means clustering to determine average voltages, for example:
[00001]$\mathrm{Averageinputvoltageof PSU}_i = \frac{.\mathrm{Math.}_{k=1}^{n} \mathrm{IV}_k^i}{t}$; and $\mathrm{Averageoutputvoltageof PDUoutlet}_j = \frac{.\mathrm{Math.}_{k=1}^{n} \mathrm{OV}_k^j}{t}$.
[0058] At step **610**, for each outlet **404**, management agent **506** may determine a power difference between such outlet **404** and each PSU **110** in the same voltage group as such outlet **404** as determined at step **608**. For example, in some embodiments, such power difference may be calculated as a power difference “distance” D.sub.i,j based on the power vectors collected at step **604**, as given by:
[00002]$D_{i,j} = .\mathrm{Math.}_{k=1}^{n}\ .\mathrm{Math.}\ \mathrm{IP}_k^i - \mathrm{OP}_k^j\ .\mathrm{Math.}$
[0059] At step **612**, for each outlet **404**, management agent **506** may map such outlet **404** to the PSU **110** having the smallest power difference (e.g., smallest power difference distance) of all PSUs **110** in the same voltage group as outlet **404**.
[0060] At the conclusion of step **612**, method **600** may end.
[0061] Method **600** may be implemented using m **100**, components thereof or any other system operable to implement method **600**. In certain embodiments, method **600** may be implemented partially or fully in software and/or firmware embodied in computer-readable media.
[0062] FIG. **7** illustrates a flow chart of an example method **700** for verification of mapping of PSUs **110** to power distribution units **104** and outlets **404** thereof, in accordance with embodiments of the present disclosure. According to certain embodiments, method **700** may begin at step **702**. As noted above, teachings of the present disclosure may be implemented in a variety of configurations of system **100**. As such, the preferred initialization point for method **700** and the order of the steps comprising method **700** may depend on the implementation chosen.
[0063] At step **702**, management agent **506** may, for a particular mapping between a PSU **110** and a particular outlet **404** (e.g., as determined by execution of method **600**), apply a power cap to all information handling systems **102** except for the information handling system **102** having the particular PSU **110**.
[0064] At step **704**, management agent **506** may cause the information handling system **102** having the particular PSU **110** to experience a change in power consumption. For example, in some embodiments, when management agent **506** has host access to the information handling system **102** having the particular PSU **110**, management agent **506** may cause a processor on the information handling system **102** to execute a predefined workload. As another example, when management agent **506** lacks host access to the information handling system **102** having the particular PSU **110**, management agent **506** may (e.g., via management controller **206** of the information handling system **102**) cause an increase in power consumption by increasing a fan speed within information handling system **102**.
[0065] At step **706**, management agent **506** may determine if the outlet **404** mapped to the particular PSU **110** of the information handling system **102** has experienced a change in power delivery commensurate with the change in power consumption of the information handling system **102**. If the outlet **404** experiences a change in power delivery commensurate with the change in power consumption of the information handling system **102**, method **700** may proceed to step **710**. Otherwise, method **700** may proceed to step **708**.

[0066] At step **708**, responsive to the outlet **404** failing to experience a change in power delivery commensurate with the change in power consumption of the information handling system **102**, management agent **506** may take a remedial action. Such remedial action may include any suitable action including without limitation communicating an alert (e.g., to a user) regarding the incorrect mapping and/or repeating method **600** to re-map outlets **404** to PSUs **110**. After completion of step **708**, method **700** may end.
[0067] At step **710**, management agent **506** may determine if all mappings of outlets **404** to PSUs **110** have been verified. If all PSUs **110** have been so verified, method **700** may end. Otherwise, method **700** may proceed again to step **702**, and steps **702** through **706** may repeat for each mapping. After completion of step **710**, method **700** may end.
[0068] Although FIG. **7** discloses a particular number of steps to be taken with respect to method **700**, method **700** may be executed with greater or fewer steps than those depicted in FIG. **7**. In addition, although FIG. **7** discloses a certain order of steps to be taken with respect to method **700**, the steps comprising method **700** may be completed in any suitable order.
[0069] As used herein, when two or more elements are referred to as “coupled” to one another, such term indicates that such two or more elements are in electronic communication or mechanical communication, as applicable, whether connected indirectly or directly, with or without intervening elements.
[0070] This disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments herein that a person having ordinary skill in the art would comprehend. Similarly, where appropriate, the appended claims encompass all changes, substitutions, variations, alterations, and modifications to the example embodiments herein that a person having ordinary skill in the art would comprehend. Moreover, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, or component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Accordingly, modifications, additions, or omissions may be made to the systems, apparatuses, and methods described herein without departing from the scope of the disclosure. For example, the components of the systems and apparatuses may be integrated or separated. Moreover, the operations of the systems and apparatuses disclosed herein may be performed by more, fewer, or other components and the methods described may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order. As used in this document, “each” refers to each member of a set or each member of a subset of a set.
[0071] Although exemplary embodiments are illustrated in the figures and described above, the principles of the present disclosure may be implemented using any number of techniques, whether currently known or not. The present disclosure should in no way be limited to the exemplary implementations and techniques illustrated in the figures and described above.
[0072] Unless otherwise specifically noted, articles depicted in the figures are not necessarily drawn to scale.
[0073] All examples and conditional language recited herein are intended for pedagogical objects to aid the reader in understanding the disclosure and the concepts contributed by the inventor to furthering the art, and are construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present disclosure have been described in detail, it should be understood that various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the disclosure.
[0074] Although specific advantages have been enumerated above, various embodiments may include some, none, or all of the enumerated advantages. Additionally, other technical advantages may become readily apparent to one of ordinary skill in the art after review of the foregoing figures

and description.
[0075] To aid the Patent Office and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants wish to note that they do not intend any of the appended claims or claim elements to invoke 35 U.S.C. § 112 (f) unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. An information handling system comprising: a processor; and a management agent configured to: collect input power information and input voltage information associated with a plurality of power supply units; collect output power information and output voltage information associated with a plurality of outlets; and for each outlet of the plurality of outlets: determine a power difference between such outlet and each of the plurality of power supply units; and map such outlet to a power supply unit of the plurality of power supply units for which the power difference between such outlet and such power supply unit is smallest among all of the power differences between such outlet and each of the plurality of power supply units.
2. The information handling system of claim 1, wherein the management agent is further configured to, based on the input voltage information and the output voltage information, select the plurality of power supply units from a larger plurality of power supply units and select the plurality of outlets from a larger plurality of outlets such that each of the plurality of power supply units selected has an input voltage approximately equal to that of each of the plurality of outlets selected.
3. The information handling system of claim 1, wherein: the input power information associated with the plurality of power supply units comprises, for each power supply unit of the plurality of power supply units, an input power vector defining an amount of input power of such power supply unit at a plurality of different times; and the output power information associated with the plurality of outlets comprises, for each outlet of the plurality of outlets, an output power vector defining an amount of output power of such outlet at the plurality of different times.
4. The information handling system of claim 3, wherein determining the power difference between a particular outlet of the plurality of outlets and each of the plurality of power supply units comprises determining a vector difference distance between the particular outlet and each of the plurality of power supply units.
5. The information handling system of claim 4, the management agent further configured to map the particular outlet to a respective power supply unit of the plurality of power supply units for which the vector difference distance between such outlet and such power supply unit is smallest among all of the vector difference distances between such outlet and each of the plurality of power supply units.
6. The information handling system of claim 1, wherein the management agent is further configured to verify a mapping between a particular power supply unit of the plurality of power supply units and a particular outlet of the plurality of outlets by: causing a change in power consumption of a system comprising the particular power supply unit; and detecting if a commensurate change occurs in power delivery of the outlet.
7. A method comprising: collecting input power information and input voltage information associated with a plurality of power supply units; collecting output power information and output voltage information associated with a plurality of outlets; and for each outlet of the plurality of outlets: determining a power difference between such outlet and each of the plurality of power supply units; and mapping such outlet to a power supply unit of the plurality of power supply units for which the power difference between such outlet and such power supply unit is smallest among all of the power differences between such outlet and each of the plurality of power supply units.
8. The method of claim 7, further comprising, based on the input voltage information and the output voltage information, selecting the plurality of power supply units from a larger plurality of

power supply units and selecting the plurality of outlets from a larger plurality of outlets such that each of the plurality of power supply units selected has an input voltage approximately equal to that of each of the plurality of outlets selected.

9. The method of claim 7, wherein: the input power information associated with the plurality of power supply units comprises, for each power supply unit of the plurality of power supply units, an input power vector defining an amount of input power of such power supply unit at a plurality of different times; and the output power information associated with the plurality of outlets comprises, for each outlet of the plurality of outlets, an output power vector defining an amount of output power of such outlet at the plurality of different times.

10. The method of claim 9, wherein determining the power difference between a particular outlet of the plurality of outlets and each of the plurality of power supply units comprises determining a vector difference distance between the particular outlet and each of the plurality of power supply units.

11. The method of claim 10, further comprising mapping the particular outlet to a respective power supply unit of the plurality of power supply units for which the vector difference distance between such outlet and such power supply unit is smallest among all of the vector difference distances between such outlet and each of the plurality of power supply units.

12. The method of claim 7, further comprising verifying a mapping between a particular power supply unit of the plurality of power supply units and a particular outlet of the plurality of outlets by: causing a change in power consumption of a system comprising the particular power supply unit; and detecting if a commensurate change occurs in power delivery of the outlet.

13. An article of manufacture comprising: a non-transitory computer-readable medium; and computer-executable instructions carried on the computer-readable medium, the instructions readable by a processor, the instructions, when read and executed, for causing the processor to: collect input power information and input voltage information associated with a plurality of power supply units; collect output power information and output voltage information associated with a plurality of outlets; and for each outlet of the plurality of outlets: determine a power difference between such outlet and each of the plurality of power supply units; and map such outlet to a power supply unit of the plurality of power supply units for which the power difference between such outlet and such power supply unit is smallest among all of the power differences between such outlet and each of the plurality of power supply units.

14. The article of claim 13, the instructions for further causing the processor to, based on the input voltage information and the output voltage information, select the plurality of power supply units from a larger plurality of power supply units and select the plurality of outlets from a larger plurality of outlets such that each of the plurality of power supply units selected has an input voltage approximately equal to that of each of the plurality of outlets selected.

15. The article of claim 13, wherein: the input power information associated with the plurality of power supply units comprises, for each power supply unit of the plurality of power supply units, an input power vector defining an amount of input power of such power supply unit at a plurality of different times; and the output power information associated with the plurality of outlets comprises, for each outlet of the plurality of outlets, an output power vector defining an amount of output power of such outlet at the plurality of different times.

16. The article of claim 15, wherein determining the power difference between a particular outlet of the plurality of outlets and each of the plurality of power supply units comprises determining a vector difference distance between the particular outlet and each of the plurality of power supply units.

17. The article of claim 16, the instruction for further causing the processor to map the particular outlet to a respective power supply unit of the plurality of power supply units for which the vector difference distance between such outlet and such power supply unit is smallest among all of the vector difference distances between such outlet and each of the plurality of power supply units.

18. The article of claim 13, the instructions for further causing the processor to verify a mapping between a particular power supply unit of the plurality of power supply units and a particular outlet of the plurality of outlets by: causing a change in power consumption of a system comprising the particular power supply unit; and detecting if a commensurate change occurs in power delivery of the outlet.